

Qualification and Approval Guide

Redbird Flight Simulations, Inc.

Basic Aviation Training Device

Model TD BASIC

TABLE-MOUNTED FLIGHT TRAINING DEVICE

Configuration: Generic, Single Engine, Land, Fixed Landing Gear, Fixed Pitch Propeller Airplane

Details: Generic G1000-style glass cockpit
Standard yoke
Vernier style engine controls

REDBIRD FLIGHT SIMULATIONS, INC.
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Redbird Flight Simulations, Inc. is requesting FAA certification of a Basic Aviation Training Device (BATD). This device meets or exceeds the criteria outlined in the FAA Advisory Circular 61-136 Appendix B.

In the interest of clarity, accuracy and brevity, we have cited the relevant portions of the Advisory Circular 61-136 detailing the requirement for a BATD.

Those portions are shown in blue, underlined and in italics, as here.

Followed by our explanation of compliance with those requirements

shown in bold and surrounded by a box, as here.

Primary, relevant components of the TD BASIC:

Qty	Manufacturer	Model/Version	Description/Function
1	Redbird Flight Simulations, Inc.	TD BASIC	Central Processor - Sim Platform Software
1	Microsoft	Windows7	Operating System
1	Microsoft	ESP	Simulation Engine
1	Redbird	Instructor Station	Map, Track, and Glideslope tracking
1	27" Industry Standard	Generic	Flat Panel exterior views & instruments
2	19" Industry Standard	Generic	Flat Panel displays for virtual instruments
1	Redbird Flight Simulations, Inc.	Redbird TD	Instrument Panel and display cabinet
1	CH Products Corporation.	Pro Pedals	Rudder Pedals
1	Redbird Flight Simulations, Inc.	RBTDY-STD	Yoke Control & Throttle Quadrant
1	SolidTEK, Inc.	USB	Keyboard/Mouse
1	Redbird Flight Simulations, Inc.	RBTD-G001	Virtual Instrument Controls Overlay
1	Redbird Flight Simulations, Inc.	RB-Switch-TD	Lower Panel Switch block

Appendix B: Section 4: COMPATIBILITY OF SOFTWARE AND HARDWARE COMPONENTS.
b. In some cases, the hardware manufacturer and the software developer do not work together in developing the ATD. Instead, the software is "licensed for use" to the ATD manufacturer and incorporated into the device. In those cases, the manufacturer must attest in writing that all hardware technical requirements (analog and digital input values) are compatible with the software used in the ATD. To do so, the manufacturer should obtain a "compatibility statement" from the software developer, which may, at the FAA's discretion, be used to satisfy this requirement. The following is an example "compatibility statement":
c. Only the owner or co-developer can validate certification of the transport delay time stated in paragraph 4b and the correct analog and digital inputs necessary to ensure that the software performs in accordance with the source code demands. Similarly, the software developer must determine the minimum computer requirements to effectively run the software.

"This is to certify that Microsoft Corporation, the owner and developer of the ESP Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software."

Appendix B: Section8: BASIC ATD DESIGN CRITERIA
a. Controls. *A Basic ATD must provide certain physical controls and may provide some virtual controls.*
(1) Physical flight and aircraft system controls should be recognizable as to their function and how they are to be manipulated solely from their appearance. Physical flight and aircraft system controls eliminate the use of interfaces such as a keyboard, mouse, or gaming joystick to control the represented aircraft model in simulated flight.

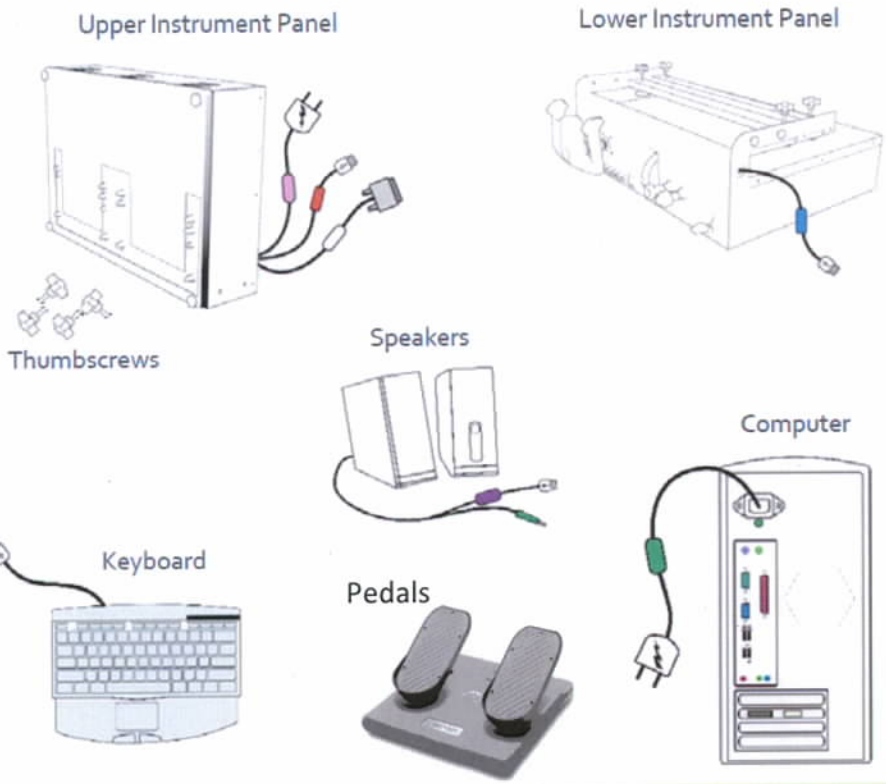
Meets Requirement(s)

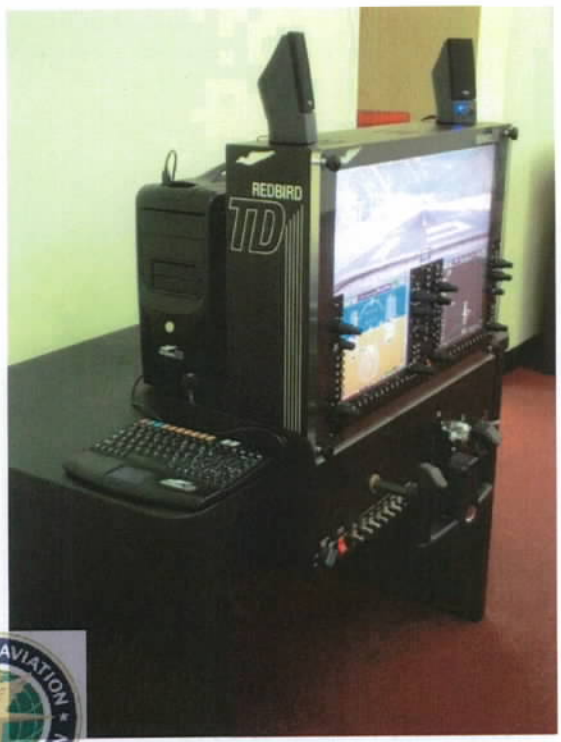
The keyboard/mouse device, supplied with the Redbird TD, is used for setting flight location and flight conditions before a flight simulation sessions begins. Once the Model TD BASIC is initialized and a training session has begun, the keyboard is used only to display the pilot's performance, show backup instruments, pause flight, end flight, introduce aircraft systems failures, and weather events. The pilot will manipulate instruments and avionics presented on the virtual instrument displays by use of a physical switch or dial correctly located on a clear panel overlaying the displays

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Model TD BASIC EXTERIOR COMPONENTS

The REDBIRD TD BASIC is an all-metal single-place flight training device. The TD BASIC is a table mounted, all metal flight training device designed to be ergonomically correct for cockpit simulation. The unit operates in any climate controlled, office type, conditions. As a rule, wherever people are comfortable, the TD BASIC and it's systems are comfortable.





The Redbird TD BASIC mounts under the table, instead of sitting on top, to put the controls in the proper position.



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TD BASIC INSTRUMENT PANEL



While the flight instruments may be virtual, the knobs, buttons, and switches that control them are quite real. Each is positioned in the correct location over the virtual instrument or device that it controls, imbedded in the a clear acrylic panel overlying the LCD screens.





AVIONICS CONTROLS WHERE THEY'RE SUPPOSED TO BE

The gauges and avionics may be virtual, but the buttons and dials that control them are real and placed exactly where you'd expect them to be thanks to Redbird's exclusive transparent overlay panel. For example, the heading bug control is exactly where you'd expect it, over the heading indicator. The same is true for every other device on the virtual panel.





Appendix B: Section8: BASIC ATD DESIGN CRITERIA

a. Controls. A Basic ATD must provide certain physical controls and may provide some virtual controls.

(2) For the purposes of this AC, virtual control is any input device to control aspects of the simulation (such as setting aircraft configuration, location, and weather) and to program, pause, or freeze the device. Virtual controls should be primarily for the instructor's use.

Meets Requirement(s)

The keyboard/mouse device, supplied with the Redbird TD, used for setting flight location and flight conditions before a flight simulation sessions begins. Once the Model TD BASIC is initialized and a training session has begun, the keyboard is used only to display the pilot's performance, show backup instruments, pause flight, end flight, introduce aircraft systems failures, and weather events. The pilot will manipulate instruments and avionics presented on the virtual instrument displays by use of a physical switch or dial correctly located on a clear panel overlaying the displays All flight controls are easily recognizable based on their position and appearance. The pilot will manipulate instruments and avionics presented on the virtual instrument display monitor by use of a physical switch/dial correctly located on a clear panel overlaying the monitor.

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Appendix B: Section8: BASIC ATD DESIGN CRITERIA
a. Controls. A Basic ATD must provide certain physical controls and may provide some virtual controls.

(3) Except for setup and/or fault mode entry, neither the keyboard nor the mouse may be used to set or position any feature of the Basic ATD in the represented aircraft for the maneuvers or flight training to be accomplished. See the control requirements listed below for necessary equipment as applicable to the aircraft model represented. The pilot must operate the additional equipment needed in order to accomplish a training procedure as listed in this appendix in the same manner in which it would be operated in the represented aircraft. For example: landing gear, wing flaps, cowl flaps, carburetor heat control, and mixture, propeller, and throttle controls.

Meets Requirement(s)

During a flight training session, the student operates all flight controls in the same manner as the actual aircraft being simulated. A flat panel monitor represents all panel instrumentation. Any required student manipulations of instruments is accomplished with physical switches/dials in the same manner as the actual instrument device.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA
a. Controls. A Basic ATD must provide certain physical controls and may provide some virtual controls.

(4) The physical arrangement, appearance, and operation of controls, instruments, and switches required by this appendix should model at least one aircraft in the family of aircraft represented as closely as practicable. Manufacturers are expected to use their best efforts to recreate the appearance, arrangement, operation, and function of realistically placed physical switches and other required controls representative of a generic aircraft instrument panel that includes at least the following:

- *Master/Battery*
- *Magnetos for each engine*
- *Alternators for each engine*
- *Fuel Boost Pumps for each engine*
- *Avionics Master*
- *Pitot Heat*
- *Rotating Beacon/Strobe, Navigation, Taxi, and Landing Lights*

Meets Requirement(s)

The switches and controls listed above closely represent the configuration, look/feel of the intended simulated aircraft.



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Appendix B: Section8: BASIC ATD DESIGN CRITERIA
a. Controls. A Basic ATD must provide certain physical controls and may provide some virtual controls.

(5) When an FAA-approved Basic ATD is in use, only the software package evaluated and approved by the FAA may be loaded for use to avoid negative impact on available system resources. This may require a separate user profile for ATD operation.

Meets Requirement(s)

During the start-up of the SD / FMX 1000 simulator, the PC configuration and file inventory is examined to ensure only tested software is installed. In the event additional files are found, an initialization error will be displayed and logging of flight training data to the student key will NOT be allowed.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA
b. Control Requirements.
Physical flight and aircraft system controls must be provided as follows:

(1) A self-centering displacement yoke or control stick that allows continuous adjustment of pitch and bank;
(2) Self-centering rudder pedals that allow continuous adjustment of yaw;
(3) Throttle or power control(s) that allows continuous movement from idle to full power settings;
(4) Mixture/condition, propeller, and throttle/power control(s) as applicable to the aircraft or family of aircraft represented; and
(5) Controls for the following items, as applicable to the airplane or family of airplanes represented:

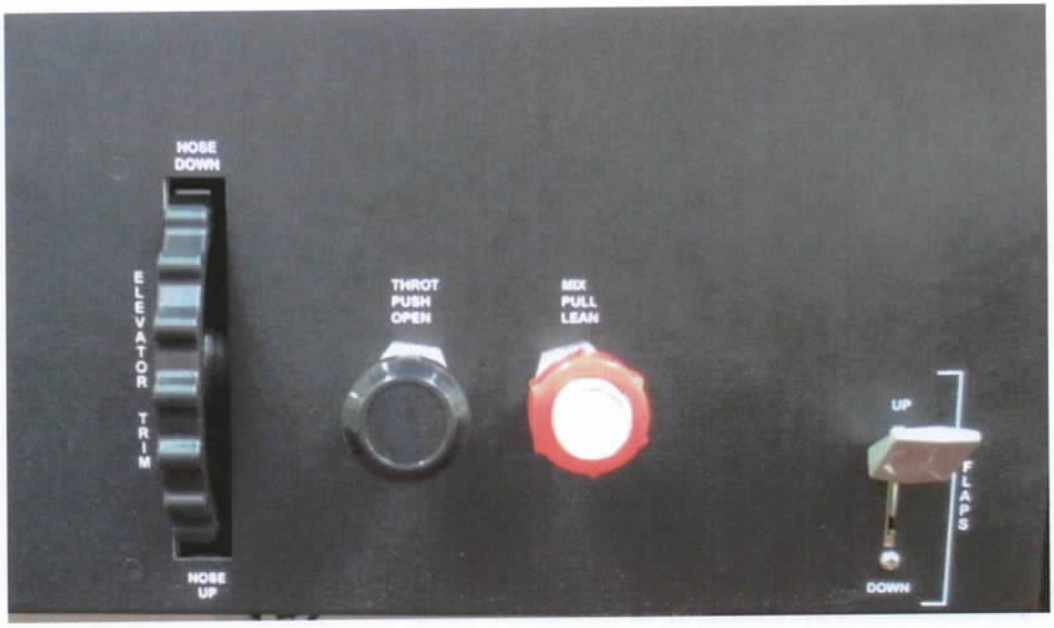
- *Wing flaps*
- *Pitch trim*
- *Communication and navigation radios*
- *Clock or timer*
- *Gear handle (N/A)*
- *Transponder*
- *Altimeter*
- *Microphone with push-to-talk switch (N/A)*
- *Carburetor heat (N/A)*
- *Cowl flaps (N/A)*

Meets Requirement(s)

The yoke and rudder pedals are self-centering devices and meet the requirements stated above. Mixture, and power provide full range control from idle to full-power and have position indicators provided on virtual display to indicate actual position. All other controls applicable for the aircraft being represented and listed above are represented either by physical device or virtual device on the LCD displays. If simulated on the displays, any requirement to manipulate the device is accomplished with a physical dial/switch similar to actual aircraft and in the proper location.

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TRIM WHEEL, POWER QUADRANT, FLAP SWITCH



Appendix B: Section8: BASIC ATD DESIGN CRITERIA
c. Control inputs

(1) Time from control input to recognizable system response (transport delay) must be 300 milliseconds or less. The manufacturer listed in the approval guide submitted for FAA approval must certify to this standard.

Meets Requirement(s)

The industry standard USB 2.0 serial connection has a raw data rate at 480Mbps and is the only connection used for all TD BASIC input devices. Each TD BASIC device is polled by software interface 200 times per/second to ensure device is both responding and meets the requirement specified above.

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Appendix B: Section8: BASIC ATD DESIGN CRITERIA

c. Control inputs

(2) The control inputs must be tested by the computer and software at each start-up and displayed as a confirmation message or a warning message that the transport delay time or any design parameter is out of original tolerances. This test should consider the items listed under Display Requirements in paragraph d below.

Meets Requirement(s)

Each control input is tested during the system start-up and the required transport delay is verified using internal start/stop timers. If any device fails the above test(s), a warning message is displayed and the simulator will stop, otherwise; a confirmation indicating "successful test condition" occurs when the Main user interface screen appears



Appendix B: Section8: BASIC ATD DESIGN CRITERIA

***d**. Display Requirements.*

(1) Instruments and Indicators as appropriate and applicable to the aircraft or family of aircraft represented:

- *Flight instruments in a standard configuration, represented either as separate traditional "round" flight instruments, or as an electronic primary flight instrument display with reversionary and back-up flight instruments*
- *A sensitive altimeter with incremental markings each 20 feet or less, operable throughout the normal operating range of the aircraft or family of aircraft represented*
- *A magnetic compass which displays incremental markings typical of that shown in the aircraft or family of aircraft represented*
- *A heading indicator with incremental markings each 5° or less, displayed on a 360° circle. Arc segments of less than 360° may be selectively displayed if desired or required, as applicable to the aircraft or family of aircraft represented*
- *An airspeed indicator with incremental markings as shown on the aircraft or family of aircraft represented; airspeed markings of less than 40 knots need not be displayed*
- *A vertical speed indicator with incremental markings each 100 feet per minute (fpm) for both climb and descent, for the first 1,000 fpm of climb and descent, and at each 500 fpm climb and descent for the remainder of a minimum ± 2,000 fpm total display, or as applicable to the aircraft or family of aircraft being represented*
- *A gyroscopic rate-of-turn indicator or equivalent with appropriate markings for a rate of 3° per second turn for left and right turns. If a turn and bank indicator is used, the 3° per second rate index must be inside of the maximum deflection of the indicator*

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• A slip and skid indicator with coordination information displayed in the conventional skid ball format where a coordinated flight condition is indicated with the ball in the center position. A split image triangle indication may be used if applicable to the aircraft or family of aircraft being represented

• An attitude indicator with incremental markings each 5o of pitch or less, from 20o pitch up to 40o pitch down or as applicable to the aircraft or family of aircraft represented. Bank angles must be identified at "wings level" and at 10, 20, 30, and
60o of bank (with an optional additional identification at 45o) in left and right banks

• Engine instruments as applicable to the aircraft or family of aircraft being represented, providing markings for normal ranges and minimum and maximum limits

• A suction gauge or instrument pressure gauge, as applicable, with a display applicable to the aircraft or family of aircraft represented

• A flap setting indicator that displays the current flap setting. Setting indications should be typical of that found in an actual aircraft

• A pitch trim indicator with a display that shows zero trim and appropriate indices of airplane nose down and airplane nose up trim, as would be found in an aircraft

• Communication radio(s) with display(s) of the radio frequency in use

• Navigation radio(s) capable of replicating both precision and non-precision instrument, including approach procedures (each with an aural identification feature), and a marker beacon receiver. For example, an Instrument Landing System (ILS), Nondirectional Beacon (NDB), Global Positioning System (GPS), localizer (LOC) or Very High Frequency Omnidirectional Range (VOR). Graduated markings as indicated below must be present on each indication as applicable. The marking should include:

  o One-half dot or less for course/glide slope deviation (i.e., VOR, LOC or ILS)

  o 5° or less for bearing deviation for Automatic Direction Finder (ADF) and radio magnetic indicator (RMI), as applicable

  o A clock with incremental markings for each minute and second, or a timer with a display of minutes and seconds

  o A magnetic compass with incremental markings each 10° or less. The compass must display the proper lead or lag during turns

  o A transponder panel that displays the current transponder setting, and

  o A fuel quantity indicator(s) that displays the fuel remaining, either in analog or digital format, as appropriate for the aircraft or family of aircraft represented.

Meets Requirement(s)

All of the above conditions are met as standard outputs from Microsoft ESP Simulation Engine software. In the conditions where the above instruments/gauges are manipulated with physical controls, the appropriate control position is represented on the instrument/gauges on the monitor.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA

d. Display Requirements.

(2) All instrument displays listed above must be visible during all flight operations. The update rate of all displays must provide an image of the instrument that:

- Does not appear to be out of focus or illegible
- Does not appear to "jump" or "step" to a distracting degree during operation, and
- Does not appear with distracting jagged lines or edges.



Meets Requirement(s)

The clarity of instruments displayed is of very high quality by using flat panel LCD display instead of projections. All instruments are focused, do NOT jump –or- step and is not distracting.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA

d. Display Requirements.

(3) Display update must be at a frequency of 10 Hz or higher. Each display must sense a change and react at a value less than the stated. Display updates must display all changes (within the total range of the replicated instrument) that are equal to or greater than the values stated below:

- *Airspeed indicator: Change of 5 knots*
- *Attitude indicator: Change of 2° in pitch and bank*
- *Altimeter: Change of 10 feet*
- *Turn and bank: Change of ¼ standard rate turn*
- *Heading indicator: Change of 2°*
- *Vertical Speed Indicator (VSI): Change of 100 fpm*
- *Tachometer: Change of 25 rpm or 2% of turbine speed*
- *VOR/ILS: Change of 1° for VOR or ¼ of 1° for ILS*
- *ADF: Change of 2°*
- *GPS: Change as appropriate for the model of GPS-based navigator represented*
- *Clock or timer: Change of 1 second.*

Meets Requirement(s)

The TD BASIC exceeds the performance requirements listed above. Display updates exceed 60Hz.

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Appendix B: Section8: BASIC ATD DESIGN CRITERIA

d. Display Requirements.

(4) Displays must reflect dynamic behavior of an actual aircraft display; e.g., a VSI reading of 500 fpm must reflect a corresponding movement in altimeter, and an increase in power must reflect an increase in the rpm indication or power indicator.

Meets Requirement(s)

The TD BASIC exceeds the performance requirements listed above.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA

e. Flight Dynamics Requirements.

(1)Flight dynamics of the ATD should be comparable to the way the represented training aircraft performs and handles. However, there is no requirement for an ATD to have control loading to exactly replicate any particular aircraft. An air data-handling package is not required for determination of forces to simulate during the manufacturing process.

Meets Requirement(s)

The simulated aircraft flight dynamics are comparable to the way actual aircraft performs and handles.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA

e. Flight Dynamics Requirements.

(2)Aircraft performance parameters (such as maximum speed, cruise speed, stall speed, maximum climb rate) should be comparable to the aircraft or family of aircraft being represented.

Meets Requirement(s)

All simulated aircraft flight dynamics are comparable to the way actual aircraft performs and handles.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA

e. Flight Dynamics Requirements.

(3)Aircraft vertical lift component must change as a function of bank, comparable to the way the aircraft or family of aircraft being represented performs and handles.

Meets Requirement(s)

All simulated aircraft flight dynamics are comparable to the way actual aircraft performs and handles.

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Appendix B: Section8: BASIC ATD DESIGN CRITERIA

e. Flight Dynamics Requirements.

(4) Changes in flap setting, slat setting, gear position, collective control or cyclic control must be accompanied by changes in flight dynamics, comparable to the way the aircraft or family of aircraft represented performs and handles.

Meets Requirement(s)

All simulated aircraft flight dynamics are comparable to the way actual aircraft performs and handles.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA

e. Flight Dynamics Requirements.

(5) The presence and intensity of wind and turbulence must be reflected in the handling and performance qualities of the simulated aircraft and should be comparable to the way the aircraft or family of aircraft represented performs and handles.

Meets Requirement(s)

All simulated aircraft flight dynamics are comparable to the way actual aircraft performs and handles.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA

f. Instructional Management Requirements.

(1)The instructor must be able to pause/freeze the system at any point for the purpose of administering instruction regarding the task.

Meets Requirement(s)

The instructor has the ability to pause/freeze the flight/system at any time using labeled function keys on the TD BASIC keyboard.



Appendix B: Section8: BASIC ATD DESIGN CRITERIA

f. Instructional Management Requirements.

(2) If a training session begins with the "aircraft in the air" and ready for the performance of a particular procedural task, the instructor must be able to manipulate the following system parameters independently of the simulation:

- *Aircraft geographic location*
- *Aircraft heading*
- *Aircraft airspeed*
- *Aircraft altitude*
- *Wind direction, speed, and turbulence*

Meets Requirement(s)

The above listed instructor related aircraft parameters can be manipulated independent of the simulation.

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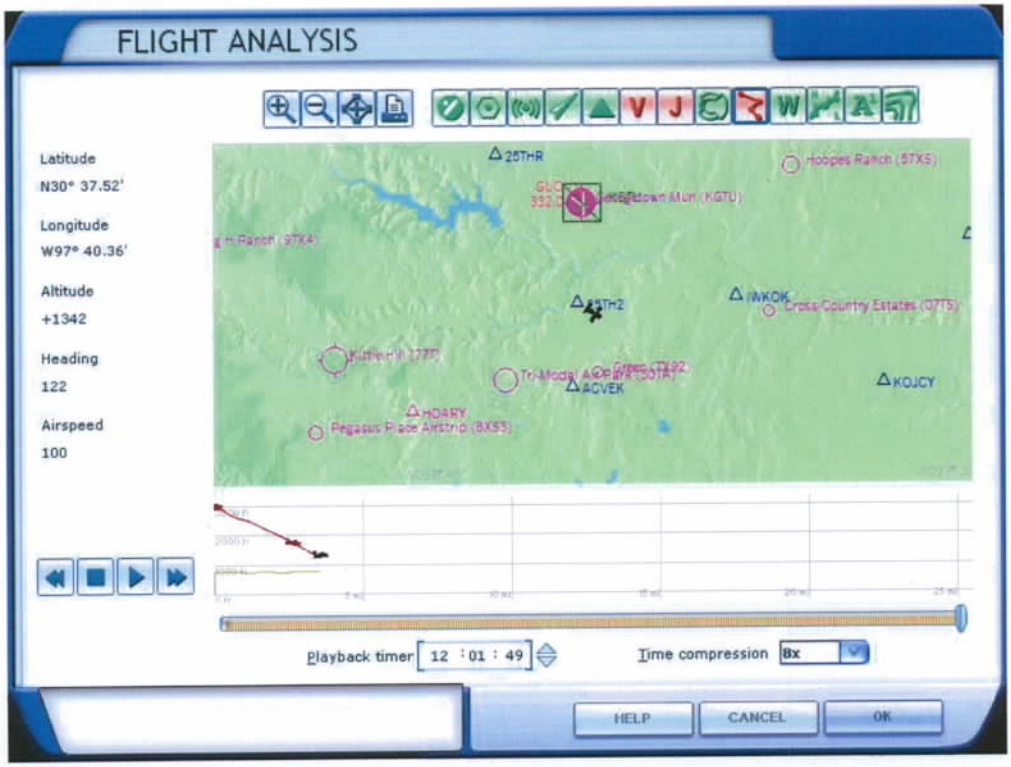
Appendix B: Section8: BASIC ATD DESIGN CRITERIA

f. Instructional Management Requirements.

(3)The system must be capable of recording both a horizontal and vertical track of aircraft movement for later playback and review.

Meets Requirement(s)

The system is capable of displaying and recording both horizontal and vertical track of aircraft movement for immediate or later playback and review. Viewing horizontal and vertical track is accomplished by pressing the ANALYZE FLIGHT key on the keyboard . By pressing this key once, the active flight is paused and an analysis screen appears depicting the aircraft's track and elevation. By pressing the key a second time, the flight resumes where it left off.



By pressing the ANALYZE Key on the keyboard, the flight is paused and this screen is displayed for the student.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA

f. Instructional Management Requirements.

(4) The instructor must be able to disable any of the instruments prior to or during a training session, and be able to simulate failure of any of the instruments without stopping or freezing the simulation to affect the failure.

Meets Requirement(s)

The instructor is able to disable instruments prior to or during a training session, and is able to simulate failure of the instruments without stopping or freezing the simulation to affect the failure.

Appendix B: Section8: BASIC ATD DESIGN CRITERIA

f. Instructional Management Requirements.

(5) The ATD must have at least a navigational area database that is local to the training facility to allow reinforcement of procedures learned during actual flight in that area. All navigational data must be based on procedures as published in part 97.

Meets Requirement(s)

The TD BASIC simulator includes a navigational data base that covers the entire world and is based on procedures as published in part 97. Navigation data is presented on the Redbird 1000 PFD and MFD which closely approximates a Garmin 1000 panel



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SPECIAL NOTE: In addition to it's qualifications as a BATD, the REDBIRD TD BASIC meets the following Advanced ATD performance criteria.

Appendix C: Section 4: FUNCTION AND SUBJECTIVE TEST CRITERIA.

Devices eligible as Advanced ATDs for creditable use should conform to an acceptable aircraft cockpit/instrument panel arrangement. The simulated systems and subsystems should be able to perform operational functions and performance maneuvers that closely mimic the represented aircraft. Specific attention should be given to ergonomic and human factors such as those shown above. These devices must be clearly eligible to receive a consensus evaluation for consideration of approval and added creditable use based on technical merit of design and function that mentally challenges the pilot's performance in both procedural and operational performance PTS tasks as if it were a real aircraft. The criteria listed in paragraph 3 above and the graphical outline checklist shown in Table 1 will be used to help determine whether the advanced design and performance of the simulation device merits FAA approval as an Advanced ATD under this publication. The FAA CET will use the checklist in Table 1 during the evaluation of an Advanced ATD.

1. Functions and Maneuvers	
a. Preparation for Flight	
(1) Preflight	X
b. Pre-Takeoff	
(1) Engine start	X
(2) Brake operation	X
c. Takeoff	
AIRPLANE Takeoff	
(i) Power plant checks	X
(ii) Acceleration characteristics	X
(iii) Nose wheel and rudder steering	X
(iv) Effect of crosswind	X
(v) Instrument	X
(vi) Landing gear, wing flap operation	X
d. In-flight Operation	
AIRPLANE In-flight Operation	
(1) Climb	
(i) Normal	X
(ii) One engine inoperative procedures	X
(2) Cruise	
(i) Performance characteristics (speed vs. power)	X
(ii) Normal and steep turns	X
(iii) Performance turns	X
(iv) Approach to stalls, i.e., stall warning (cruise, takeoff/approach, and landing configuration)	X
(v) High angle of attack maneuvers (cruise, takeoff/approach, and landing)	X
(vi) In-flight engine shutdown	X
(vii) In-flight engine restart	X
(3) Descent	
(i) Normal	X
(ii) Maximum rate	X

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e. Approaches	
(1) Non-precision	X
(i) All engines operating	X
(ii) One or more engines inoperative	X
(iii) Approach procedures (NDB, VOR, DME Arc, LOC/BC, LOC, LDA, SDF, ASR, LNAV/VNAV, GPS & LPV, and additional approach types when developed and certified)	X
(iv) Missed approach (all engines operational, one or more engines inoperative)	X
(2) Precision	
(i) PAR – normal	X
(ii) ILS	X
(iii) GPS-WAAS	
(iv) GPS-LAAS	
(v) Effects of crosswind	X
(vi) With engine(s) inoperative	X
(vii) Missed approach	
(A) Normal	X
(B) With engine(s) inoperative	X
(C) From steep glide slope	X
f. Surface Operations (Airplane - Post Landing)	
(1) Landing roll	X
(2) Braking operation	X
(3) Reverse thrust operation, if applicable	X
g. Any Flight Phase	
(1) Aircraft and power plant systems	
(i) Electrical	X
(ii) Flaps (**airplane**)	X
(iii) Fuel and oil	X
(iv) Landing gear (**airplane**)	X
(2) Flight management and guidance systems	
(i) Automatic pilot	X
(ii) Flight director/system displays	X
(iii) Navigation systems	X
(iv) Stall warning/avoidance (**airplane**)	X
(v) Multi function displays	X
(vi) Primary flight displays	X
(3) Airborne procedures	
(i) Holding	X
(4) Engine shutdown and parking	
(i) Systems operation	X
(ii) Parking brake operation (**airplane**)	X

Meets Requirement(s)

The TD BASIC aircraft configurations represents the appropriate cockpit and instrument panel layout. The simulated systems and subsystems allow for the above operational functions and performance maneuvers to be performed.